SheepFormers: Vision Transformer-Based Sheep Breed Classification on a Small Dataset

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Extended Abstract:

Research Area

Knowing a sheep's breed matters for breeding plans, productivity and farm profit, but visual inspection, body measurements and DNA tests are subjective, slow or expensive. Earlier CNN and ensemble studies mostly used small datasets and stopped at reporting accuracy, with no tool a farmer could use. This work asks whether a Vision Transformer (ViT), usually thought to need huge datasets, can still classify sheep breeds from only 1,680 images. It builds on our earlier ensemble study on the same dataset [1].

Goals/Objectives

The objectives are to show that a fine-tuned ViT can reach state-of-the-art accuracy on a small sheep-face dataset, to compare ViT variants with earlier CNN and ensemble results through tuning and ablation, and to deploy the best model in a real-time mobile app for farmers, buyers and sellers.

Methodology

The dataset holds 1,680 face images of four breeds (Marino, Poll Dorset, Suffolk and White Suffolk), 420 each, resized to 224×224 and split 80/10/10 into training, validation and test sets. We fine-tuned six variants (ViT-Google with patch size 32, ViT-MAE, ViT-VAN, ViT-BEiT, ViT-ResNet50 and ViT-DiT), tuning epochs, batch size, learning rate, optimizer and loss; the final setting was 100 epochs, batch size 32, learning rate 0.00002, Adam and categorical cross-entropy. We also ran ablations on augmentation, positional encoding and patch size, 5-fold cross-validation, and a test on 80 external images. The best model was converted to TensorFlow Lite and built into a Flutter mobile app (Fig. 1).

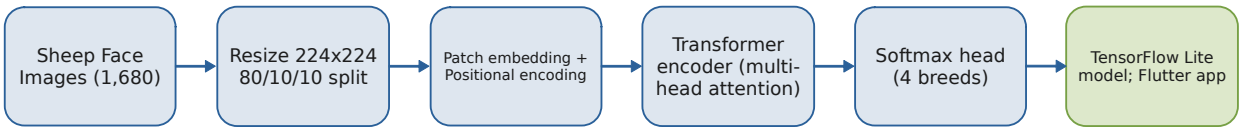


Figure 1.Overall pipeline of the proposed SheepFormers system, from image preparation to the mobile app.

Results and Analysis

ViT-Google gave the best results, with 98.21% test accuracy and an AUC of 0.9987 (Table I), making only three errors on 168 test images. This beats our earlier ensemble model (97.92%), EfficientNet (97.62%) and the CNN baselines we trained (best 84.7%). Five-fold cross-validation gave a mean accuracy of 0.9817 ± 0.0007, and the model scored 97.5% on 80 external images. Augmentation gave no extra gain, while removing positional encoding or using patch size 16 lowered accuracy; ViT-DiT and ViT-ResNet50 struggled most with Marino and White Suffolk. The main limits are the small, controlled dataset and the lack of tests under real farm conditions such as poor lighting, occlusion and thick wool. Future work will add more varied data, federated learning, lighter ViT models and usability studies with farmers.

Table 1.Test accuracy and AUC of the ViT variants

Model	ViT-Google	ViT-MAE	ViT-VAN	ViT-BEiT	ViT-ResNet50	ViT-DiT
Accuracy (%)	98.21	97.62	95.24	84.52	83.93	79.17
AUC	0.9987	0.9962	0.98	0.9341	0.9075	0.85

Keywords: *vision transformer, sheep breed classification, small dataset, smart farming, image classification*

References

[1] G. M. S. Himel, M. M. Islam, and M. Rahaman, "Vision intelligence for smart sheep farming: Applying ensemble learning to detect sheep breeds," Artif. Intell. Agric., vol. 11, 2024, doi: 10.1016/j.aiia.2023.11.002.

[2] G. M. S. Himel, M. M. Islam, and M. Rahaman, "SheepFormers: Vision transformer-based sheep breed classification on small datasets for smart farming," J. Electr. Syst. Inf. Technol., vol. 13, Art. no. 20, 2026, doi: 10.1186/s43067-026-00326-0.

[3] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," arXiv:2010.11929, 2020.